

Extrema and the Mean Value Theorem

ANSWER KEY



Critical points and extrema

- Find the critical points of $f(x) = 2x^3 - 3x^2 - 12x + 5$.
 $f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2) = 6(x - 2)(x + 1)$. Critical points: $x = 2$ and $x = -1$.
- Find the absolute maximum and minimum of $f(x) = x^3 - 3x$ on $[-2, 3]$, checking critical points and endpoints.
 $f'(x) = 3x^2 - 3 = 0$ gives $x = \pm 1$ (both in $[-2, 3]$). Values: $f(-2) = -2$, $f(-1) = 2$, $f(1) = -2$, $f(3) = 18$. Absolute max: $f(3) = 18$. Absolute min: $f(-2) = f(1) = -2$ (tied).

The Extreme Value Theorem

- State the Extreme Value Theorem and explain why it guarantees $f(x) = x^2 - 4x$ has an absolute max and min on $[0, 5]$.
The EVT states that a continuous function on a closed interval $[a, b]$ attains both an absolute maximum and an absolute minimum on that interval. Since $f(x) = x^2 - 4x$ is continuous (a polynomial) and $[0, 5]$ is closed, the EVT guarantees both extrema exist.

Rolle's Theorem

- Verify that Rolle's Theorem applies to $f(x) = x^2 - 4x + 3$ on $[1, 3]$, and find the value(s) of c guaranteed by the theorem.
 $f(1) = 0$ and $f(3) = 0$: equal endpoint values, and f is continuous and differentiable everywhere, so Rolle's Theorem applies. $f'(x) = 2x - 4 = 0$ gives $c = 2$, which lies in $(1, 3)$.
- Explain why Rolle's Theorem does NOT apply to $f(x) = |x - 2|$ on $[0, 4]$, even though $f(0) = f(4) = 2$.
 f is not differentiable at $x = 2$ (it has a corner), so one of Rolle's Theorem's hypotheses (differentiability on the open interval) fails — the theorem's guarantee does not apply.

The Mean Value Theorem

- For $f(x) = x^3$ on $[0, 2]$, find the value(s) of c guaranteed by the Mean Value Theorem.
Average rate of change: $\frac{f(2) - f(0)}{2 - 0} = \frac{8}{2} = 4$. Need $f'(c) = 3c^2 = 4$: $c^2 = \frac{4}{3}$, $c = \frac{2}{\sqrt{3}} \approx 1.15$ (taking the positive root within $[0, 2]$).

- 7 A car travels 180 km in 2 hours. Using the Mean Value Theorem, explain why the car must have been traveling at exactly 90 km/h at some instant.
- Average velocity $= \frac{180}{2} = 90$ km/h. Since position is a differentiable function of time, the MVT guarantees some instant c where the instantaneous velocity $v(c) = 90$ km/h exactly.
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Using the MVT to bound function values

- 8 Given $f(2) = 5$ and $-3 \leq f'(x) \leq 4$ for all x , use the MVT to find the range of possible values for $f(6)$.
- By the MVT, $f(6) - f(2) = f'(c)(6 - 2) = 4f'(c)$ for some $c \in (2, 6)$. Since $-3 \leq f'(c) \leq 4$:
 $-12 \leq f(6) - 5 \leq 16$, so $-7 \leq f(6) \leq 21$.
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Consequences of the MVT: monotonicity

- 9 Explain, using the MVT, why $f'(x) > 0$ on an interval implies f is strictly increasing on that interval.
- For any $a < b$ in the interval, the MVT gives $\frac{f(b) - f(a)}{b - a} = f'(c)$ for some $c \in (a, b)$. Since $f'(c) > 0$ and $b - a > 0$, we get $f(b) - f(a) > 0$, i.e. $f(b) > f(a)$ — so f is strictly increasing.
- 10 Explain why two functions f and g with $f'(x) = g'(x)$ for all x must differ by a constant, using the MVT.
- Let $h(x) = f(x) - g(x)$. Then $h'(x) = f'(x) - g'(x) = 0$ for all x . If h were not constant, the MVT applied between any two points where h differs would give $h'(c) \neq 0$ for some c , a contradiction. So $h(x) = C$ for some constant C , i.e. $f(x) = g(x) + C$.
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Visualizing the Mean Value Theorem

- 11 The graph shows $f(x) = x^3$ on $[0, 2]$ together with the secant line through the endpoints and the tangent line at $x = \frac{2}{\sqrt{3}}$ (found earlier).
- The tangent line at $c = \frac{2}{\sqrt{3}}$ is parallel to the secant line connecting $(0, 0)$ and $(2, 8)$, both having slope 4 — this is exactly the geometric statement of the MVT: somewhere between the endpoints, the instantaneous slope matches the average slope.
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Absolute extrema on a closed interval — a capstone problem

- 12 Find the absolute maximum and minimum of $f(x) = x^4 - 4x^2 + 3$ on $[-2, 2]$, checking all critical points and endpoints.
- $f'(x) = 4x^3 - 8x = 4x(x^2 - 2) = 0$ gives $x = 0, \pm\sqrt{2}$, all within $[-2, 2]$. Values:
 $f(-2) = 16 - 16 + 3 = 3$, $f(-\sqrt{2}) = 4 - 8 + 3 = -1$, $f(0) = 3$, $f(\sqrt{2}) = -1$, $f(2) = 3$. Absolute max: 3 (at $x = -2, 0, 2$). Absolute min: -1 (at $x = \pm\sqrt{2}$).
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Rolle's Theorem with trigonometric functions

- 13** Verify that Rolle's Theorem applies to $f(x) = \sin x$ on $[0, \pi]$, and find the value of c guaranteed by the theorem.
- $f(0) = 0$ and $f(\pi) = 0$: equal endpoint values, and $\sin x$ is continuous and differentiable everywhere, so Rolle's Theorem applies. $f'(x) = \cos x = 0$ gives $c = \frac{\pi}{2}$, which lies in $(0, \pi)$.
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An applied Mean Value Theorem problem

- 14** A runner completes a 10 km race in 40 minutes. Using the Mean Value Theorem, find the runner's average speed, and explain why there must be some instant during the race when the runner's instantaneous speed equals this average.
- Average speed $= \frac{10}{40} = 0.25$ km/min $= 15$ km/h (distance divided by time). Since distance is a differentiable function of time, the MVT guarantees some instant c during the race where the instantaneous speed equals exactly 15 km/h.